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Past, Present, and Future of Citation Practices in HCI

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Past, Present, and Future of Citation Practices in HCI

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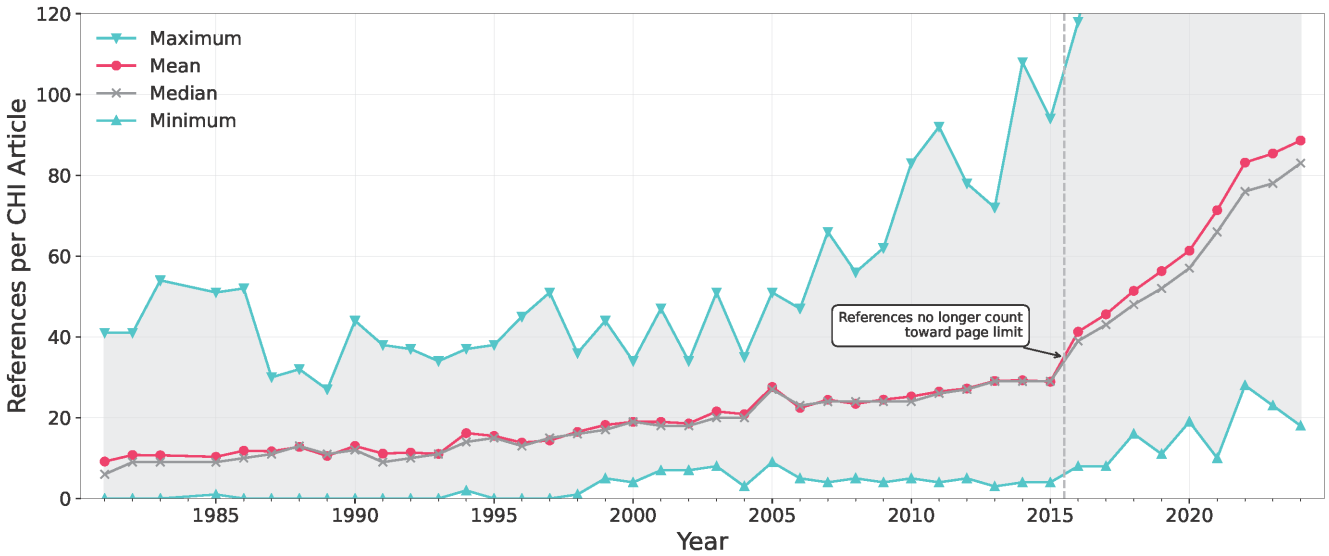

Figure 1: Number of references per article in the Proceedings of the ACM CHI Conference from 1981 to 2024.

Abstract

Science is a complex system comprised of many scientists who individually make decisions that, due to the size and nature of the academic system, largely do not affect the system as a whole. However, certain decisions at the meso-level of research communities, such as the Human-Computer Interaction (HCI) community, may result in deep and long-lasting behavioral changes in scientists.

In this article, we provide empirical evidence on how a change in editorial policies introduced at the ACM CHI Conference in 2016 destabilized the CHI research community and launched it on an expansive path, denoted by a year-by-year increase in the mean number of references included in CHI articles. If this near-linear trend continues undisrupted, an article at CHI 2030 will include *on average* almost 130 references.

The trend toward more citations reflects a citation culture where quantity is prioritized over quality, contributing to both author and peer reviewer fatigue. Our exploratory analysis highlights the profound impact of meso-level policy adjustments on the evolution of scientific fields and disciplines, urging all stakeholders to carefully consider the broader implications of such changes.

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CCS Concepts

• **Human-centered computing** → *Human computer interaction (HCI)*; • **General and reference** → *Surveys and overviews.*

Keywords

references, citations, CHI, bibliometric analysis, event study, meta-science, meta-HCI

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1 Introduction

Citation serves as the backbone of academic rigor, enabling researchers to build upon previous work, acknowledge contributions, and weave a rich tapestry of interdisciplinary knowledge. Citation practices, as a tangible expression of scholarly discourse, provide valuable insights into the evolving norms and values of the research community [28]. Editorial policies shape the landscape of academic publishing, influencing not only the structure and content of the research published but also the citation practices of researchers. Understanding the impact of policy decisions on citation practices is crucial for ensuring that scholarly communication serves both the authors and the broader field effectively.

In this work, we examine how the citation practices of researchers in the Human-Computer Interaction (HCI) community have changed after an editorial policy decision introduced in 2016. In this year, the page restrictions at HCI's top conference, the ACM Conference on Human Factors in Computing Systems (CHI), were lifted. This policy decision had a profound impact on the citation practices of the CHI community. We first explore the historic development of the number of references in CHI articles:

RQ1: *How has the number of citations from CHI articles to other works developed in the CHI Proceedings?*

Our analysis of 11,542 articles published at the ACM CHI Conference between 1981 and 2024 provides empirical evidence of a year-by-year increase in the mean number of references included in CHI articles. This trend was enabled by the change in policies in 2016, allowing authors to include an unlimited number of references in their articles. The subsequent change in the community's citation practices signifies a departure from previous patterns and suggests the community's entry into a novel trajectory (cf. Figure 1). If this trend continues undisrupted, articles at CHI 2030 will include *on average* about 130 references. Such a scenario poses practical challenges for authors and peer reviewers alike, straining the authors' capacity to meaningfully engage with the literature and the peer reviewers' capacity to thoroughly vet the cited literature during the peer review process.

We explore a number of potential reasons for the observed increase in the mean number of references included in CHI articles:

RQ2: *What other factors, beside the 2016 policy decision, could potentially contribute to the observed growth in references, and how have these factors evolved over the years?*

We plot and and visually analyze a number of co-occurring trends. As part of this, we also investigate whether there is a bias in awarding articles at CHI post-policy change. If articles with a high number of references were to systematically be presented with awards at the CHI Conference, this could send a signal to the community and potentially incentivize authors to include more references in their articles, contributing to the observed growth in the mean number of references per CHI article.

Last, we investigate the significance of these observed trends in shaping the expansive citation practices in the CHI community:

RQ3: *Was there a significant change in citation practices of authors at CHI after the editorial policy change in 2016? If so, how do the co-occuring factors contribute to this trend?*

We investigate this research question with an event study, demonstrating that the policy change at CHI '16 had a profound and destabilizing effect on the CHI community. The CHI community has clearly entered a different trajectory at this point in time (cf. Figure 1), marked by increasing academic pressures contributing to fatigue of both authors and peer reviewers.

The rising number of references in CHI articles raises important questions about the nature of academic discourse, the balance between comprehensiveness and focus in scholarly communication, and the sustainability of current citation practices. The challenges brought by the escalating citation practices within the HCI community are multifaceted, encompassing the expansion of the CHI Conference, the diversification of topics and methodologies, and the elevation of standards for what constitutes a publication-worthy contribution. Such developments, while indicative of progress, also contribute to the increasing complexity of the academic ecosystem, negatively impacting both authors and peer reviewers. With the volume of references expanding, the feasibility of conducting thorough and effective peer reviews is called into question. This article discusses a number of potential solutions to this trend.

We argue it is high time for the CHI community to pause, reflect, and reevaluate how the HCI field approaches the practice of citation.

In light of our findings, this paper aims to spark a conversation within the CHI community and beyond about the future of scholarly communication and academic publishing. By examining the contextual factors and ramifications of changing citation practices, we invite scholars, practitioners, and policymakers to reflect on past decisions and identify sustainable pathways forward.

2 Background and Related Work

2.1 The Field of Human-Computer Interaction

Human-Computer Interaction is both a field and a discipline [8, 26]. As a field, HCI encompasses diverse disciplines, such as computer science, cognitive science, psychology, design, sociology, anthropology, and more [8, 18, 26, 29]. HCI draws from these diverse areas to understand how humans interact with computers and how to design user-friendly systems. As an academic discipline, HCI is taught in structured curricula at many universities in dedicated degree programs, educating students in the principles and practices of HCI. Human-Computer Interaction is also recognized as a professional discipline with specific roles, such as UX researchers and interaction designers.

In the remainder of this paper, we refer to HCI as a field.

Each year, a growing number of articles are being published in the field of HCI. The international top conference in HCI is the ACM Conference on Human Factors in Computing Systems (CHI), which has been held each year since 1981 (with exception of 1984). CHI is the most prestigious conference venue for researchers in HCI, with many labs submitting their articles exclusively to only this conference venue. As such, the CHI Conference can serve as a reference standard for the entire field and discipline of HCI.

From its humble beginnings in the early 1980s, the CHI Conference has been expanding since 2005 in terms of articles accepted at the conference. Except for the COVID-19 pandemic years in 2021 and 2022, the CHI Proceedings have exhibited near linear growth since 2005. In 2024, the CHI Proceedings exceeded 1,000 articles. While this expansion is plan-driven and directed by CHI's Steering Committee, it also reflects the expansion and diversification of the wider field of HCI.

In this expansive and evolving research landscape, the practice of citing relevant works is important for HCI researchers. Citations serve as a vital link, connecting current research efforts with the rich history of past works. Citations acknowledge the contributions of other researchers and weave a rich tapestry of knowledge that informs and enriches new inquiries. For a field as interdisciplinary as HCI, where the integration of diverse perspectives and methodologies is crucial, citations provide a structured way to navigate and summarize the vast array of existing knowledge for readers without deep subject-matter knowledge and expertise. Further, citations

enhance the credibility and rigor of academic discourse, enabling scholars to build upon a verified body of work. Proper citation practices contribute to the integrity of the HCI field, fostering an environment of collaboration and continuous learning. As such, citations are not merely a scholarly obligation but a critical element in the academic and intellectual growth of the HCI community.

2.2 Meta-Research on the CHI Community and the Field of HCI

While the majority of HCI researchers conduct research *within* their field, some researchers have also examined aspects *about* the field of HCI. This is commonly referred to as *meta-research* [15]. Meta-research is valuable, as it enables the research community to reflect on its research practices [30, 31]. The CHI Proceedings, in particular, are a fruitful study subject for meta-scientific and bibliometric inquiries in the field of HCI, due to the size and importance of the CHI Conference for the field of HCI. Bartneck and Hu, for instance, presented a scientometric analysis of the CHI Proceedings, focusing on organizations and countries that contribute to CHI [5]. Their work highlighted the difficulty of judging quality in the context of best paper awards, finding a mismatch between awarded papers and citations received. Our paper also includes an analysis of awards, however we focus on analyzing whether there is a systematic bias in awarded papers.

Lee et al. presented a bibliometric analysis of the CHI Proceedings (from 1981 to 2018), using a citation network analysis to identify emerging topics in CHI [23]. Liu et al. also conducted a bibliometric analysis of CHI publications [26]. Their keyword co-word analysis quantified and described the thematic evolution of the HCI field based on 3,152 CHI articles published between 1994 and 2013. Like our work, their article includes a comparison between two time periods (1994–2003, 2004–2013), highlighting the underlying trends in the HCI community.

Pohl and Mottelson presented a quantitative meta-research study on paper writing at CHI, including an investigation on citations received [32]. Their analysis of 6,578 CHI papers found that readability, title length, and novelty influence citation counts. Our work differs in that we do not investigate writing and only focus on citations given, not received.

Another area of meta-research is the use of literature reviews in HCI. Stefanidi et al. conducted an analysis of the CHI Proceedings and ACM Transactions on Computer-Human Interaction (TOCHI) including 189 articles [35]. Like in our work, the authors note an "insufficient consensus of what to expect of literature reviews in HCI." The diverse types and methodologies of literature reviews in HCI prompt us to split our analysis of literature reviews into a broad and narrow part.

Finally, Kaye presented a quantitative analysis of CHI, examining author counts, gender, and repeat authorship [17]. Among other findings, the work identified a trend toward an increasing number of authors per paper. We identify the same trend in our work, and similar to Kaye, we aim to encourage a discussion about what the preferred state of CHI should be.

3 Method

In this paper, we employ both exploratory analysis (with descriptive statistics and visual analysis) and statistical analysis with linear regression models and an event study specification. The research involved gathering, quantifying, and analyzing data from the ACM Digital Library (ACM–DL), as detailed in the following section.

3.1 Data Collection

The ACM Conference on Human Factors in Computing Systems (CHI) is the HCI field's largest and most important annual conference. Therefore, the Proceedings of the CHI Conference are a proxy for investigations into the wider field of HCI. In particular, the digital proceedings are a fertile ground for investigating the citation practices of HCI authors. The CHI Proceedings are stored in ACM's Digital Library, which is the main source of data in our research.

We collected all articles published in the CHI Proceedings from 1981 to 2024 (with exception of 1984) from the ACM–DL. We focused on full research articles and excluded articles in the companion proceedings or extended abstracts. We further excluded session details, panel sessions, keynotes, and abstract-only entries using regular expressions. For each article, we collected the full set of references included in the article and the set of authors from the ACM–DL. Further, we downloaded the PDF document and collected the full reference section from the ACM–DL for each article. This resulted in a large dataset of CHI articles ($N = 11,542$), with their respective full set of references and authors.

3.2 Analyzing the Mean Number of References in CHI Articles (RQ1)

To answer RQ1, we count the number of references in each CHI article. For each year in the CHI Proceedings, we calculate the mean, median, minimum, and maximum number of references. Visual inspection yields the insight that a clear change in slope occurred after the year 2015 (cf. Figure 1).

Simple linear regression analysis was conducted to evaluate the extent to which Year could predict the mean and minimum number of References after the policy change (2016–2024). We report the regression equation for each model, R^2 values explaining the variance in the models, t-values and p-values for the slope coefficient, and the error metrics MAE (mean absolute error) and MSE (mean squared error). The t-values for slope quantify how many standard errors the estimated slope coefficient (β_1) is away from zero. It is used to test the null hypothesis that the slope is zero, which would imply no linear relationship between the predictor (Year) and the response variable (References). F-tests are conducted testing the overall significance of the regression models, reporting F-statistic and p-value. The two linear regression models are then used to predict the mean and minimum number of citations in future CHI proceedings.

The mean number of references per CHI article is the key variable in our work. In the following section, we describe how we visualized and analyzed a number of other contextual factors potentially playing a role in shaping the citation practices of the CHI community.

3.3 Exploring Other Contextual Factors (RQ2)

To answer RQ2, we explore the historical development of a number of co-occurring trends.

3.3.1 Analyzing bias in awarding articles. We examine whether a systematic bias can be observed in awarded papers in past proceedings of the CHI Conference. As award, we consider both "best paper" awards and "honorable mentions." We count the references in awarded and unawarded articles at the CHI Conference. We then statically analyze the difference in the mean number of references in articles that received an award ($M_{awarded}$) and the mean number of references in articles that did not receive an award ($M_{regular}$) for each proceedings year from 2007 to 2024. We report the t-test statistic, p-values, 95% confidence intervals, level of significance, and effect size (using Cohen's *d*).

3.3.2 Collaboration and co-authorship at CHI. For each article in the dataset, we count the number of authors. We plot the development of co-authorship over time and visually analyze the correlation of growth trends in authors and publications at CHI.

3.3.3 Literature reviews. Literature reviews aim to capture the current state of a research area or field. Given the rising number of scholarly articles (at CHI and globally), we expect that CHI authors are increasingly conducting literature surveys. To study this, we identified literature reviews by searching each CHI article's full text (excluding the references section) for the keywords "literature review," "literature survey," and "systematic literature review." We manually verified each occurrence, ensuring that only articles explicitly stating having conducted a literature review were included. Note that CHI authors, particularly in earlier proceedings, often referred to the related work section as a literature review. These instances are included in our plots. Additionally, some authors claim to have conducted an "extensive" or "exhaustive" literature review without detailing the process. Our data encompasses "literature review" in a broad sense, while "systematic literature reviews" are represented more narrowly, typically indicating articles that adhere to a more rigorous scientific process. We visualize the results, testing our assumption of a rise in the absolute and relative number of literature reviews at CHI.

3.3.4 Citations to pre-print articles. The landscape of academic publishing is changing. We examine how CHI authors cite pre-print articles. Pre-print articles are versions of scholarly articles that precede publication in peer-reviewed journals and conference venues. While traditionally shunned upon due to not having undergone formal peer review, it has become common place in some research fields, such as Machine Learning, to cite pre-print articles. One of the most popular pre-print archives is arXiv, launched by Cornell University in 1991. In recent years, an exponentially growing number of articles are being submitted to this pre-print archive. The high number of available pre-prints and the success of pre-prints in other fields may motivate CHI authors to also include more citations to pre-prints in their CHI articles.

To investigate this, we identified the number of citations to arXiv pre-prints in the references of each article in the CHI Proceedings by counting the number of references including a mention of "arxiv." For counting the number of occurrences of arXiv in the references section, we used a case-insensitive regular expression that accounts for hyphenation within the word arXiv due to potential line breaks. We then plot the absolute and relative number of citations to arXiv pre-prints for each proceedings year and visually analyze the results.

3.3.5 Citations to datasets and software code repositories. With the rise of data-driven research methodologies and machine learning, CHI papers may have started referencing more data sources, scientific software and tools, code repositories, and prior data-heavy studies, which could be another potential factor contributing to the increase in the mean number of references in CHI articles. We investigate this trend by counting the number of references to five popular open data and source code repositories:

- *osf.io*, a cloud-based management solution for open science, often used to share datasets and other data related to HCI studies,
- *zenodo.org*, an open repository for datasets and other research related digital artefacts,
- *github.com*, a source code repository that, with git LFS, can also store big datasets,
- *kaggle.com*, a web-based platform for sharing datasets for machine learning, and
- *huggingface.com*, a repository for storing and sharing machine learning related files.

As above, we limit our investigation to the articles' reference section. We ignore citations in footnotes within the article's full text to make the results comparable to the other findings in this article. We plot the number of citations to data and source code repositories and visually analyze the results.

3.3.6 Citations to questionable publishers. Predatory journals are questionable publications that accept articles with minimal peer review. These journals are generally regarded as having a lower quality compared to more rigorously peer-reviewed journals [7]. Without trade-off cost between references in CHI articles, CHI authors may cite articles from potential predatory journals and questionable publishers more often.

To investigate this, we compare the references included in CHI articles with entries in Beall's List of Potential Predatory Journals and Publishers [7]. We first downloaded Beall's List[1] and merged the lists of journals and publishers into one list. We removed any additional notes and entries that are generic, such as "Qualitative Research" and "American Journal," since these entries would produce many false positive matches. We then checked for exact matches between the entries in Beall's List and the references included in CHI articles (1981–2024), using basic string normalization techniques (e.g., lower-casing, removal of line breaks and special characters, and expansion of ampersand characters into the word 'and'). This comparison yielded 5,126 potential matches between entries in Beall's List and CHI references. Because the names of predatory journals are often designed to partially match with reputable journals [7], each of the 5,126 potential matches was manually reviewed to sort out false positives. We plot the number of citations to potential predatory journals and publishers and visually analyze the results.

[1] https://beallslist.net

Note that this approach is limited in several ways, pertaining to the limitations of Beall's List and the string matching approach. First, Beall's List is subjective and it should not be the only source of quality appraisal. Second, the list contains entries with notes that the journal or publisher may, in fact, not be predatory. Beall's List also mentions several instances where a previously reputable journal has been taken over by a predatory publisher. Third, while an archived version is available, Beall's List is no longer maintained. Finally, exact string matching is imperfect, although we address false positives with manual review. The presented figures can be considered a lower boundary, as there may be more matches (especially if including generic titles) that are not identified with the exact matching approach.

3.4 Analyzing the Significance of the Observed Trend (RQ3)

To answer RQ3, we analyze the impact of the policy change on the outcome variable (mean number of references in CHI articles) with an orthogonalized event study specification. An event study is a statistical method used to assess the impact of an event or intervention on an outcome by analyzing changes in the outcome variable before and after the event within a defined time window. The event of interest, here, is the 2016 policy decision to lift page restrictions on the references included in CHI articles, and the event windows are the equal-sized pre- and post-event periods (2007–2015 and 2016–2024). With the event study, we examine both the significance of the observed change in citation behavior among CHI authors and the effect of different confounders on the mean number of references in CHI articles. The rich set of covariates includes for each year t:

- Authors_t: mean number of authors per article,
- Arxiv_t: mean number of citations to arXiv pre-prints per article,
- Repos_t: mean number of citations to code and data repositories per article,
- Reviews_t: mean number of literature reviews per article,
- Predators_t: mean number of citations to potential predatory journals and questionable publishers per article, and
- AwardRatio_t: the relative prominence of awarded articles compared to unawarded articles, defined as the ratio between the mean number of references (M) in awarded and unawarded articles in a given proceedings year t: $\text{AwardRatio}_t = M_{\text{awarded},t}/M_{\text{unawarded},t}$.

The regression model for the event study can be written as:

$$\begin{aligned}\text{References}_t = \beta_0 + \beta_1\text{Post}_t + \beta_2\text{AwardRatio}_t + \beta_3\text{Authors}_t + \\ \beta_4\text{Arxiv}_t + \beta_5\text{Repos}_t + \beta_6\text{Reviews}_t + \beta_7\text{Predators}_t + \\ \beta_8(\text{Post}_t \cdot t) + \beta_t t + \epsilon_t\end{aligned}$$

with ϵ_t being an unobserved error term. Post_t is an indicator variable denoting the time period (1 if $t \geq 2016$ and 0 otherwise). The parameters β_1 and β_8 are the key coefficients of interest in this event study specification. Coefficient β_1 indicates a mean difference in References between the pre- and post-event periods. The interaction term β_8 captures a change in slope at the event point ($t = 2015.5$). The term β_t captures trends that vary linearly over time, ensuring that the model accounts for systematic changes that occur regardless of the intervention.

The time variable t is centered around its mean (the event point, $t = 2015.5$). Orthogonalization was applied to address multicollinearity among predictors, ensuring that the estimated coefficients reflect independent contributions of each variable. That is, we regressed each variable on t and use only the residuals u_t for regression:

$$\text{Variable}_t = \gamma_0 + \gamma_1 t + u_t$$

where u_t is the part of the variable not explained by t. This approach retains variability in the levels of variables and removes the temporal correlation while keeping other meaningful relationships. We report the coefficients, t-statistic for each coefficient with p-values, 95% confidence intervals, and overall significance of the model with an F-test.

4 Findings

Over the course of 43 years (from 1981 to 2024, with exception of 1984), authors at the CHI Conference cited a total of 558,080 articles. The mean number of references per CHI article over the entirety of the CHI Proceedings is 48.35 references ($SD = 35.09$, $Min = 0$, $Max = 496$). However, the field of HCI has evolved from its humble beginnings, and a more in-depth look at these descriptive statistics is warranted. In particular, we examine the increase in the mean number of references per CHI article in the following section.

4.1 The Rising Number of References in CHI Articles (RQ1)

We first provide a historical overview of citation trends in the CHI Proceedings, from the conference's beginnings to the current trend. We then extrapolate this trend into the future to provide an outlook for the year 2030. This year is not too distant (only five years from the time of writing), and we believe focusing on the year 2030 in our prediction of future trends will – if the current trend is not broken – provide an accurate and realistic estimation of the future of citations at CHI.

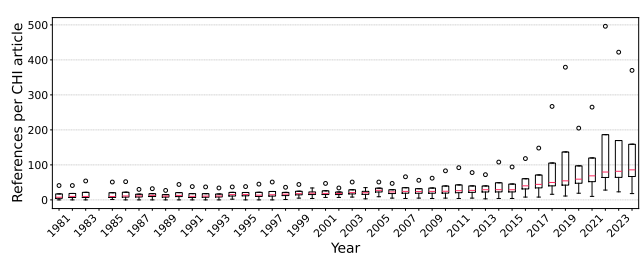


Figure 2: Number of references per article in the CHI Conference Proceedings from 1981 to 2024.

4.1.1 Historical development. In the early years of the CHI Conference, it was not uncommon for authors to submit an article without a single citation to other works. Zero-reference articles continued to be a part of the CHI Proceedings with only few exceptions until the year 1997 (see Figure 1 and Figure 2). From 1998 to 2017, the minimum number of references per article has been positive and in

2015 CfP (excerpt)

A Paper is no more than 10 pages long, while a Note is no more than 4 pages long. **This includes** figures, **references**, appendices and an abstract of less than 150 words long. Over length submissions will be rejected.

2016 CfP (excerpt)

A Paper is no more than 10 pages long, while a Note is no more than 4 pages long. **References do not count toward these lengths.** The lengths do include figures, appendices, and an abstract of less than 150 words. Submissions that exceed these limits will be rejected.

Figure 3: from the Call for Papers (CfPs) at the CHI Conference in 2015 and 2016 (own highlighting).

the one-digit range. Since 2017, the minimum number of references has increased into the double-digits.

Between 1981 and 2015, the mean number of references per CHI article showed only modest linear growth (cf. Figure 1 and Figure 2). After 2015, a clear and sudden quantitative change in the CHI community's citation practices took place. Clearly, the CHI community entered a novel trajectory at this point in time, characterized by a year-by-year increase in the mean number of references per article.[2] While there were on average 28.8 references per article in the 2015 CHI Proceedings, the number of references per CHI article more than tripled to an average of 88.6 references per CHI article in 2024.

For about two decades since its inception, the maximum number of references per CHI article hovered at around 40. After 2015, the maximum number of references skyrocketed to up to 496 references in one article in 2022. The box plots in Figure 2 demonstrate that the maximum number of references per CHI article has become more volatile in recent years. During the years of the COVID-19 pandemic (2020 and 2021), one can observe a slump in the maximum number and upper quartile of the number of references per CHI article. However, there is, on average, a clear tendency for CHI authors to include more references in their articles compared to the years before 2015 and the mean number of references included in CHI articles is monotonously growing each year.

4.1.2 Extrapolation of the observed trend. We extrapolate the observed trend using simple linear regression models (LM_1 and LM_2; see Table 1) to predict the mean and minimum number of references per article in future instances of the CHI Conference Proceedings (see Figure 4). Both models are significant ($F(1,7) = 308.96$, $p < 1e^{-6}$ and $F(1,7) = 6.81$, $p = 0.03$, respectively), with model LM_1 explaining 98% of the variance in the mean number of references. As depicted in Figure 4, the year-by-year growth in the mean number of references follows a near-linear pattern. Each year since 2016, the predicted mean number of references in CHI articles increases by approximately 6.45 references. We estimate that if the current observed trend is not broken, articles published at CHI 2030 will contain on average 129.5 references. The minimum number of references per article increases by 1.8 references per year. It is predicted to reach 33.7 references in 2030, which is more than the mean number of references from the year 2015 (28.9 references).

4.1.3 Root cause. A diverse and complex community consisting of thousands of members, such as the CHI community, is not likely to suddenly change its course, unless change is mandated from top down. In our case, an editorial policy change launched the CHI community on its expansive path. In this section, we discuss the policy change that is the root cause for the observed non-linearity in the CHI community's trajectory.

[2]Throughout this article, we refer to the mean number of references. As depicted in Figure 1, the median number of references follows a very similar trend.

Table 1: Simple linear regression models evaluate the extent to which Year (t) can predict the mean (LM_1) and minimum (LM_2) number of References (y) in CHI articles. The two models are used to predict the number of references per article in future CHI proceedings in Figure 4.

	LM_1 (Mean number of references)	LM_2 (Min number of references)
Year range	2016–2024	2016–2024
Regression equation	$y = \beta_0 + \beta_1 t$	$y = \beta_0 + \beta_1 t$
Coefficient β_0 (intercept)	-12971.92	-3620.33
Coefficient β_1 (slope)	6.45	1.80
t (for slope β_1)	$t = 17.58$	$t = 2.61$
	$p < 1e^{-6}$	$p = 0.03$
MAE	1.91	3.99
MSE	6.29	22.18
R^2	0.98	0.49
F-test	$F(1,7) = 308.96$	$F(1,7) = 6.81$
	$p < 1e^{-6}$	$p = 0.03$

The CHI Conference organizing committee (Program Chairs) in consultation with the Executive Committee of the ACM Special Interest Group on Computer-Human Interaction (SIGCHI) introduced a decisive change in the year 2016. In this year, the conference's Call for Papers (CfP) moved away from counting references toward the overall page limit of CHI articles, as highlighted in the excerpts reproduced in Figure 3. Articles published at CHI '16 were allowed to include an unlimited number of references, as long as the article's content fit the 10-page limit, with references no longer counting toward this page limit. While sub-committees were introduced as another important change in the year 2016, allowing an unlimited number of references clearly is the main cause of the increase in the mean number of references identified in our work.

Further, the "Guide to a Successful Paper or Note Submission" at CHI 2016 [1] stressed the importance of citing relevant previous work, explicitly asking *"is prior work adequately reviewed?"* The 2016 Guide also highlighted the importance of reproducibility of authors work, with *"letting others build on your work"* being regarded as *"the entire purpose of a CHI Paper"* [1]. These direct quotes from the 2016 Guide were stark reminders of the importance of citing

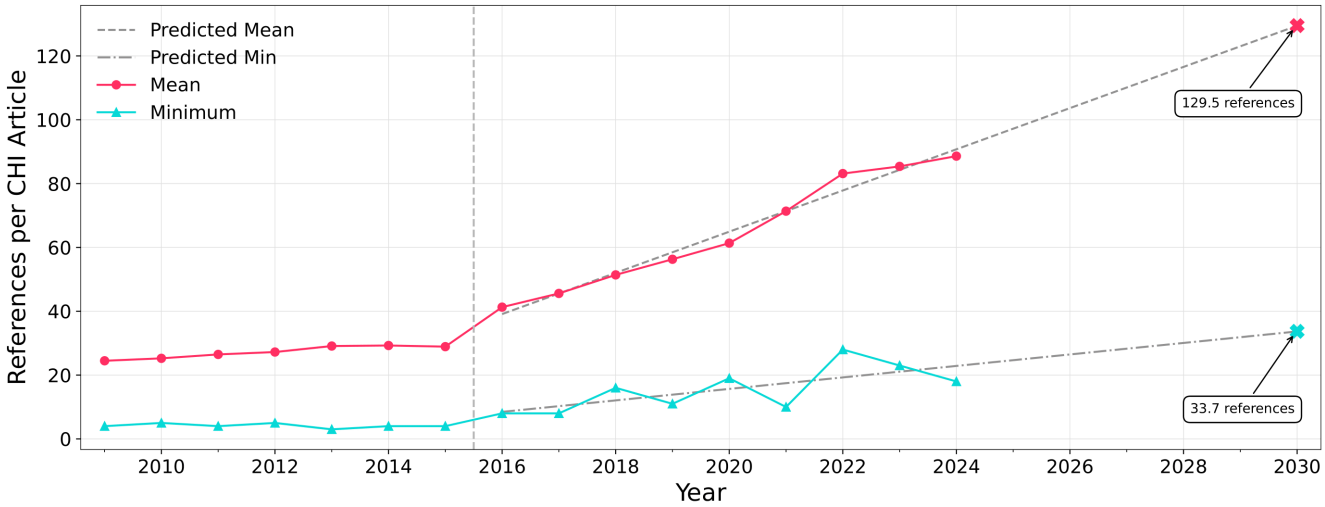


Figure 4: Extrapolation of the observed trend in the mean (red) and minimum (blue) number of references per CHI article. CHI articles are predicted to include on average almost 130 references in the year 2030. The minimum number of references per CHI article is estimated to reach 33.7 references per article in 2030, which is more than the 2015 average level of references.

prior work and may have contributed to making the community pay special attention to their references in the year 2016.

These policy changes nudged the community on its expansive path. Several other factors, such as systematic bias in awards presented to articles, may have also played a role, as described in the following section.

4.2 Additional Contextual Factors Influencing HCI Citation Practices (RQ2)

The CHI community is vast and highly diverse, and there are many factors that potentially shape and make up the citation practices of the members of the CHI community. While the editorial policy change introduced in 2016 clearly is the enabler of the rise in the number of references included in CHI articles, in this section we take a step back and explore a number of contextual factors affecting the citation practices at the CHI Conference. The six factors we investigate are a potential bias in awards (Section 4.2.1), an increase in the average number of authors (Section 4.2.2), an increase in the number of literature reviews (Section 4.2.3), an increasing number of citations to arXiv pre-print articles (Section 4.2.4), an increase in citations to data and code repositories (Section 4.2.5), and an increase in the number of citations to potential predatory journals and publishers (Section 4.2.6). These six factors are used as confounding variables in our event study specification (see Section 4.3). We do not claim that this list of factors is exhaustive. Beside the investigated changes, other observable or latent factors could contribute to the trend, as is the case with any complex system under change.

4.2.1 Are high-citation articles rewarded in the CHI community? Each year, awards are given to a select number of accepted articles at the CHI Conference. Typically, between 14.6% and 24.6% of articles are given an award at the CHI Conference (based on awards presented at CHI from 2015–2024). These awards are a signal to the community on which articles are to be considered high quality. As a quality signal, awards can motivate authors to change their behavior. If articles with a high number of references were to systematically be presented with awards, this could potentially incentivize CHI authors to include more references in their articles.

Table 2: Comparison of mean number of references in articles awarded a best paper award or honorable mention ($M_{awarded}$) with the mean number of references in unawarded articles ($M_{regular}$) in the CHI Proceedings from 2007 to 2024.

Year	$M_{awarded}$	$M_{regular}$	t	CI (95%)	d	p
2007	29.966	23.368	2.860	[2.045, 11.149]	0.580	0.005 *
2008	28.029	22.598	2.632	[1.364, 9.499]	0.491	0.009 *
2009	28.106	23.739	2.453	[0.863, 7.872]	0.393	0.015 *
2010	29.138	24.181	2.999	[1.704, 8.21]	0.420	0.003 *
2011	25.972	26.582	-0.369	[-3.859, 2.64]	-0.048	0.713
2012	27.923	27.069	0.496	[-2.534, 4.242]	0.068	0.620
2013	30.213	28.772	0.978	[-1.457, 4.339]	0.118	0.329
2014	31.398	28.66	2.036	[0.095, 5.381]	0.227	0.042 *
2015	31.219	28.153	2.298	[0.445, 5.685]	0.243	0.022 *
		Policy Change				
2016	43.470	40.705	1.616	[-0.597, 6.127]	0.170	0.107
2017	49.884	44.497	2.779	[1.581, 9.194]	0.283	0.006 *
2018	57.865	49.880	3.629	[3.665, 12.307]	0.369	0.000 *
2019	61.142	54.991	2.616	[1.534, 10.768]	0.242	0.009 *
2020	66.051	60.125	2.678	[1.582, 10.271]	0.241	0.008 *
2021	72.965	70.987	0.691	[-3.642, 7.598]	0.064	0.490
2022	87.922	81.933	1.520	[-1.747, 13.725]	0.150	0.129
2023	88.969	84.774	1.222	[-2.544, 10.934]	0.117	0.222
2024	92.958	87.651	1.795	[-0.494, 11.109]	0.144	0.073

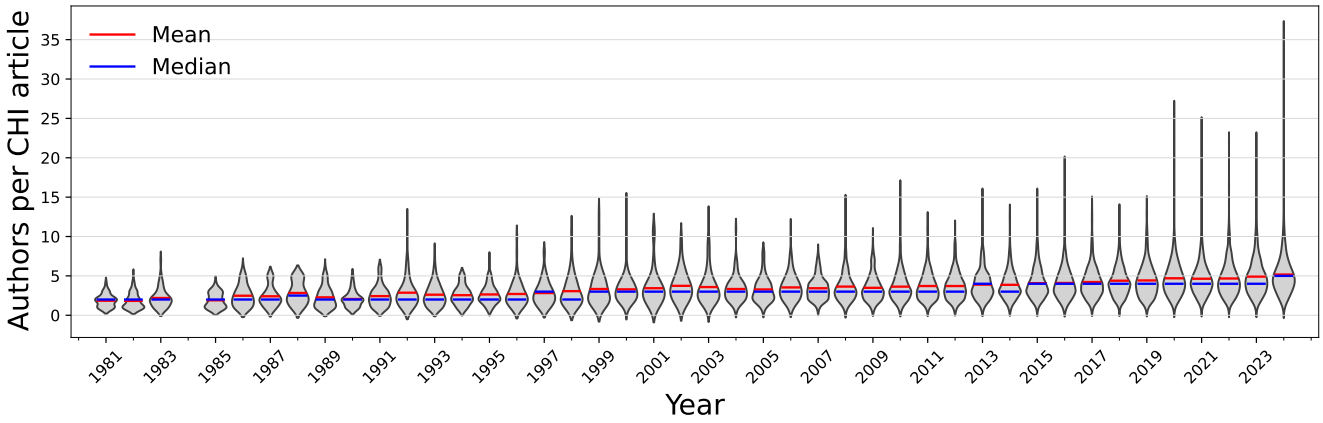


Figure 5: The number of co-authors in CHI papers has increased over time, highlighting the trend toward increased collaboration in HCI.

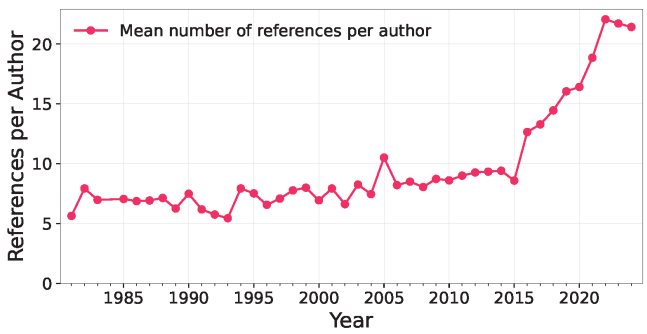


Figure 6: Mean number of references per author in CHI articles

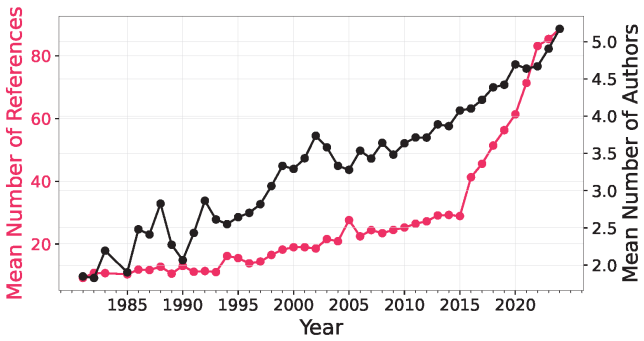


Figure 7: Mean number of references (primary axis) and mean number of authors (secondary axis) per CHI article.

Table 2 shows no evidence of a significant systematic bias toward awarding high-reference articles in recent years (2021–2024).

For instance, in the 2024 CHI Proceedings, the mean number of references for awarded articles was 92.96 compared to 87.65 for articles without award. This difference is not significant ($p > 0.05$). However, awarded articles had significantly more references than unawarded articles ($p < 0.05$; Cohen's $d = 0.23 \ldots 0.37$) in the years immediately before and after the change (2014–2020, with exception of 2016). While awards were also given to high-reference articles in some earlier periods (2007–2010), awards were systematically given to high-reference articles around the policy change (with exception of 2016). This systematic awarding of articles with above-average references may have sent a quality signal to authors, thus potentially contributing to launching the CHI community on its current expansive trajectory.

4.2.2 Collaborative research is increasing. There has been an increase in collaboration in the field of HCI leading to articles with a greater number of co-authors. This is also reflected in the CHI Proceedings. Figure 5 depicts the number of authors per CHI article for each CHI proceedings year. The mean number of authors per CHI article has increased over time, from an average of 1.85 authors per article in 1981 to an average of 5.17 authors per article in 2024. Figure 5 also demonstrates that in the years between 2020 and 2024, there have been articles surpassing 20 authors in the CHI proceedings. The CHI 2024 Proceedings contain an article with 36 authors.

Figure 6 plots the average number of references per author in CHI articles since the inception of the CHI Conference. In 1981, a CHI article included on average 5.6 references per author. This number stayed stable over three decades, with only a slight increase up to 2015. But there is a noticeable change in the mean number of references per author after 2015. Since 2016, spurred by the increase in the mean number of references per CHI article, the mean number of references per author has increased. In 2024, the mean number of references per author has reached 21.4 references per author.

Figure 7 plots the mean number of authors (right axis) against the mean number of references (left axis) per CHI article. While there clearly is a non-linearity visible in the latter time series, the former time series has been growing fairly linearly. The plot visually demonstrates that the growth in co-authorship cannot fully explain the growth in the mean number of references per CHI article.

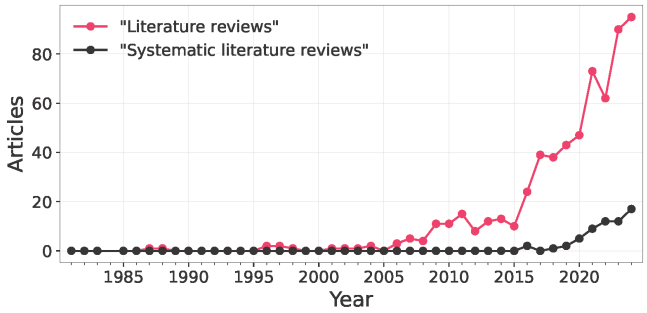


Figure 8: Absolute number of articles in the CHI proceedings reporting having conducted a literature review

Figure 9: Relative number of articles in the CHI proceedings reporting having conducted a literature review

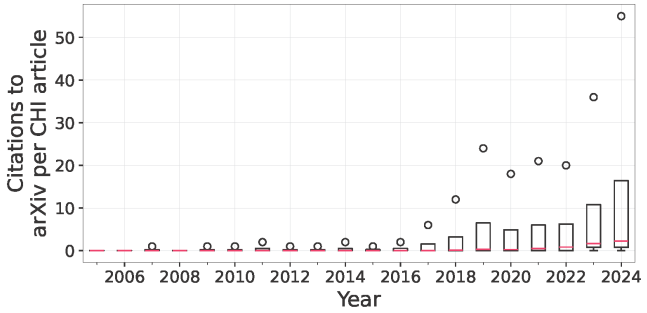


Figure 10: Formal citations to arXiv pre-print papers in CHI articles

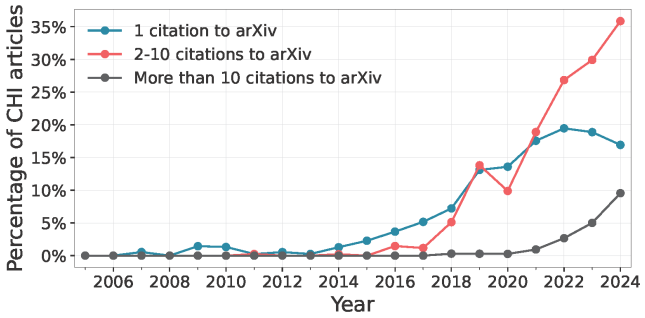


Figure 11: Relative number of articles formally citing arXiv pre-prints n times

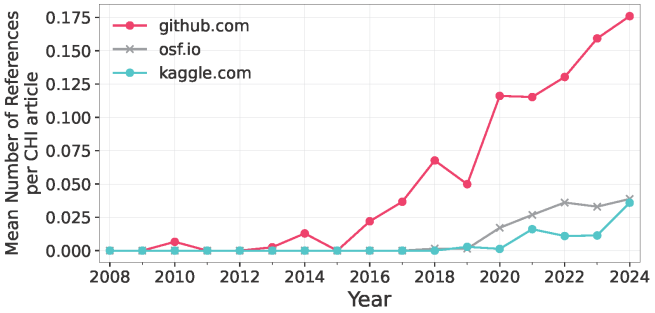


Figure 12: Mean number of citations to data and code repositories per CHI article

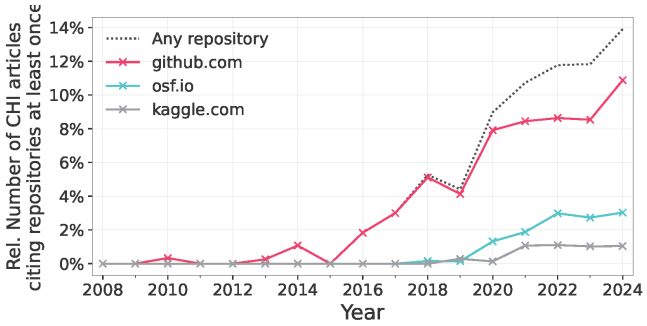
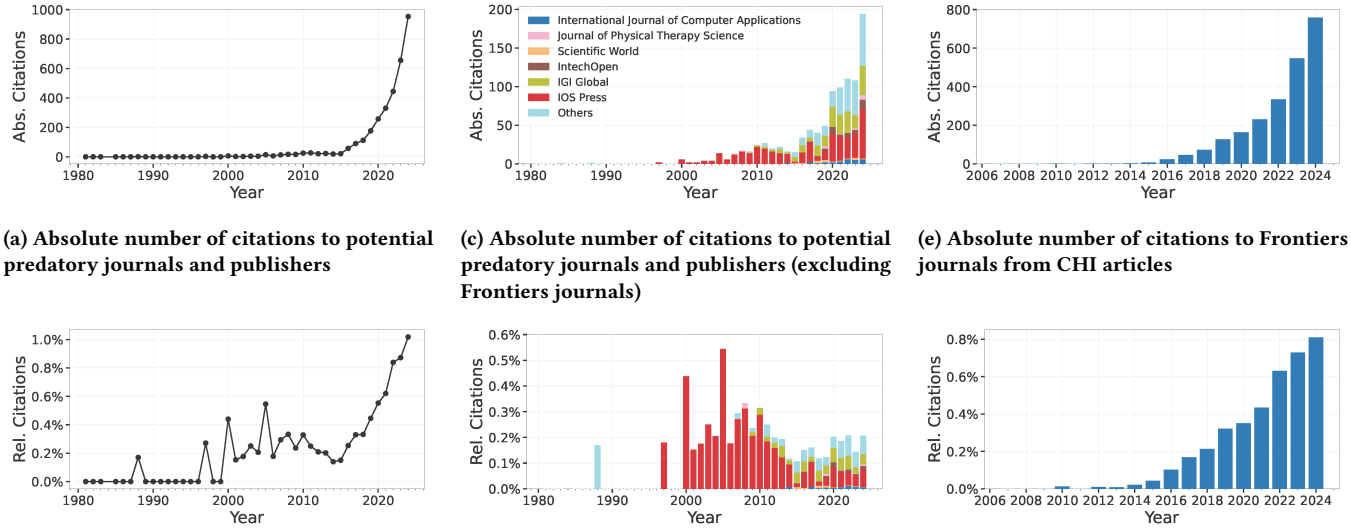


Figure 13: Relative number of CHI articles citing at least one data or code repositories

4.2.3 Literature reviews are becoming more common at CHI. In total, we identified 615 papers reporting having conducted a "literature review" and 60 systematic literature reviews in the CHI Proceedings from 1981–2024. The time series in Figure 8 showcases that since lifting the page restrictions in 2015, it has become increasingly common for CHI authors to conduct and publish literature reviews. About 10% of all articles in the latest four CHI Proceedings mention having conducted a "literature review," with only a slight drop in 2024 (see Figure 9).

Figures 8 and 9 demonstrate that systematic literature reviews in CHI are a recent and growing phenomenon. Before 2016, there was no instance of authors calling their work a systematic literature review. In recent years, it has become increasingly common for CHI authors to conduct systematic literature reviews, with up to 2% of the published articles in a given proceedings series reporting systematic literature reviews. In 2024, the CHI Proceedings included 17 systematic reviews (1.6% of the articles published in that year).



(a) Absolute number of citations to potential predatory journals and publishers

(c) Absolute number of citations to potential predatory journals and publishers (excluding Frontiers journals)

(e) Absolute number of citations to Frontiers journals from CHI articles

(b) Relative number of citations to potential predatory journals and publishers

(d) Relative number of citations to potential predatory journals and publishers (excluding Frontiers journals)

(f) Relative number of citations to Frontiers journals from CHI articles

Figure 14: Citations to potential predatory journals and publishers from CHI articles have become more common since 2016 (a and b). This growth is to a great extent produced by citations to large questionable publishers, such as Frontiers Media (e and f). While citations from CHI articles to smaller questionable publishers are growing in absolute terms (c), they remain limited in the recent decade in relative terms (d).

4.2.4 Pre-print articles are increasingly being cited by CHI authors. Even though arXiv was made available to the World Wide Web in 1993, we find that none of the CHI articles before the year 2006 included formal citations to arXiv pre-print articles. Figure 10 depicts the number of citations to arXiv pre-prints in CHI articles since 2006. We note there is a change in this time series after 2015. While the mean number of citations to arXiv pre-print articles has only slightly increased since 2016 (from 0.03 references in 2015 to 4.24 references in 2024), citations to pre-print articles have overall become more accepted among some CHI authors. One article in CHI '24 included 55 references to arXiv. Figure 11 depicts the relative number of articles citing arXiv pre-prints n times, with n split into buckets (1 citation, 2–10 citations and more than 10 citations). From this figure, one can note an increase in the number of articles that cite a high amount of arXiv pre-prints (>10 citations). In recent proceedings, over a third of the CHI papers include more than one but less than 10 citation to arXiv. Given that this change coincides with the year 2016, citations to arXiv articles are one potential factor contributing to the increase in references per CHI article.

4.2.5 Data-driven research is on the rise at the CHI Conference. Figure 12 plots the mean number of references to code and data repositories in CHI articles. While the number of formal citations to such repositories per CHI article is still low, we identify a clear increase in the number of citations to code and data repositories occurring after the year 2015. Before 2015, almost no authors formally cited these repositories, whereas today, such citations have become more common. In the proceedings of CHI 2024, 147 articles (13.9%) cite at least one data and code repository (see Figure 13). Zenodo and Huggingface were cited too infrequently to include in figures 12 and 13. GitHub was cited most often with articles (10.8%) in the CHI 2024 Proceedings including at least one reference to this repository. This empirical trend in the citation practices reflects the broader trend toward applying data-driven methodologies and machine learning in HCI research.

4.2.6 CHI authors increasingly cite questionable publishers. We identified 3,298 citations from CHI articles to potential predatory journals and publishers in the CHI Proceedings (1981 – 2024). There is a marked increase in citations to potential predatory journals and questionable publishers since 2016, both in absolute and relative numbers (see Figure 14a and Figure 14b, respectively).

Citations to Frontiers Media – the publisher which successfully pressured Jeffrey Beall to shut down his list – account for the vast majority of these citations. Splitting Frontiers from the rest of potential predatory publications, we can see that citations from CHI articles to journals published by Frontiers Media have monotonously increased since 2015, both in absolute and relative terms (see Figure 14e and Figure 14f). In 2024, every CHI paper includes on average about 0.8 references to a Frontiers journal (cf. Figure 14f). The journals from Frontiers Media cited most often in the CHI Proceedings are Frontiers in Psychology (n = 873, 39.72% of all citations to Frontiers Media), Frontiers in Human Neuroscience (n = 248, 11.28%), Frontiers in Robotics and AI (n = 213, 9.69%), Frontiers in Virtual Reality (n = 130, 5.91%), Frontiers in Neuroscience (n = 86, 3.91%), Frontiers in Psychiatry (n = 68, 3.09%), Frontiers in Public Health (n = 67, 3.05%), Frontiers in Computer

Science (n = 55, 2.5%), Frontiers in ICT (n = 37, 1.68%), Frontiers in Neurology (n = 32, 1.46%), Frontiers in Artificial Intelligence (n = 31, 1.41%), and Frontiers in Behavioral Neuroscience (n = 26, 1.18%). This is a testament that the CHI community has embraced Frontiers Media as a legitimate publisher.

Besides Frontiers Media, the remaining matches are mostly one-off citations to potential predatory journals and publishers. While there is an increase in the absolute number of citations to these potential predatory journals (other than Frontiers; see Figure 14c), the relative number of citations remained stable and limited in the recent decade (Figure 14d). Among these citations, IOS Press (n = 418) and IGI Global (n = 195) are occurring most often, followed by IntechOpen (n = 36), International Journal of Computer Applications (n = 28), Scientific World (n = 15), Journal of Physical Therapy Science (n = 13), Cambridge Scholars Publishing (n = 9), Global Media Journal (n = 8), and Research and Reviews (n = 8).

4.3 Factors Explaining the Rise in the Mean Number of References (RQ3)

Our event study provides insights into the significance of the change observed in this work and the factors influencing the mean number of references in CHI articles across the study period (see Table 3). The indicator variable for the post-event period (β_1) shows a statistically significant effect ($p = 0.006$), suggesting a notable shift in the mean number of references after the policy change. The interaction term (β_8) is also significant, indicating a significant change of slope post-event.

Among the other variables, the awards given at the CHI Conference to high-reference articles (β_2) had no statistically significant effect on the mean number of references in articles. As visually demonstrated in Figure 7, the mean number of authors (β_3) also had no significant effect on the mean number of references. Among the other coefficients, citations to potential predatory publishers (β_7) is statistically significant ($p = 0.007$), indicating that changes in predatory citation patterns may play a meaningful role in shaping citation practices among CHI authors. This can be largely attributed to the publisher Frontiers Media becoming a mainstay in the references of CHI articles since 2015. Conversely, the coefficients for citations to arXiv (β_4) and repositories (β_5), as well as literature reviews (β_6) are not statistically significant, suggesting these factors may have a limited direct impact on the mean number of references in CHI articles. The time trend coefficient (β_t) is statistically significant, pointing to other potential underlying effects explaining the trend in the mean number of references per article. These findings highlight the complexity of citation dynamics and the multifaceted factors influencing scholarly referencing practices. As mentioned in Section 4.2, this investigation is not exhaustive and there could be multiple other potential relationships that warrant exploration in future work.

5 Discussion

This article presented meta-research in the field of HCI, providing clear descriptive evidence of quantitative and qualitative changes in the citation practices of the CHI community co-occurring since the year 2016. Specifically, we have described seven co-occurring changes in citation practices:

(1) *An increase in the mean number of references included in CHI articles* (Section 4.1):
In the 2016 Call for Papers, references were no longer included in the page limit of CHI articles. This editorial decision launched the CHI community on an expansive path, characterized by a year-by-year increase in the mean number of references included in CHI articles. Our event study specification provides empirical evidence of a significant change in citation behavior, with the CHI community departing from established patterns in between the years 2015 and 2016. The top-down policy change opened the floodgates, allowing authors to cite more and more works in their articles.

(2) *Systematic bias in awarding articles post-policy change (Section 4.2.1):*
A systematic bias in awarding articles was observed. In the years following the policy change (with exception of 2016), articles with above-average number of references were systematically presented with awards. However, the event study found this systematic bias in awarding articles not to be a statistically significant factor in predicting the mean number of references in CHI articles.

(3) *An increase in collaboration and the mean number of references per author* (Section 4.2.2):
It is becoming more common for authors in HCI to collaborate and the mean number of authors per CHI article is expanding. An increase in the number of co-authors means that the set of authors may potentially contain authors from different disciplines. This could imply that authors potentially draw from a broader range of interdisciplinary sources, hence leading to more citations to articles from different fields. However, both our visual and statistical analysis found no evidence of growth in authorship being a significant factor contributing to the growth in the mean number of references in CHI articles, as also evident in Figure 6. The growth in the mean number of references exceeds the growth in authors (which has been quite linear). Author growth alone, therefore, does not serve as explanation for the increase in the mean number of references per CHI article.

(4) *An increase in the number of literature reviews* (Section 4.2.3):
The corpus of scholarly literature is expanding at a rapid rate, and we found that CHI authors increasingly conduct both systematic and unsystematic literature reviews. The editorial decision to lift the page restrictions in 2016 greatly contributed to the growing popularity of literature reviews at CHI. The years during the COVID-19 pandemic saw an acceleration of this trend, denoted by an increase in the relative number of CHI articles conducting literature reviews. This can be explained by authors not being able to conduct in-person user studies, a common method in HCI, during the pandemic. Literature reviews typically contain an above average number of references, which may contribute to the increase in the mean number of references included in CHI articles. However, as depicted in Figure 8, the trend toward literature reviews already started before the pandemic. The

Table 3: The results of an event study indicate a significant change in the mean number of references (dependent variable) per CHI article before (2007–2015) and after (2016–2024) the decision to lift page restrictions at the CHI Conference.

Statistic	Value				
R^2	0.999	$df_{regression}$		9	
Adj. R^2	0.997	$df_{residuals}$		8	
F-test	$F = 687.6, p < 1e^{-9}$				
Parameter	**Coefficient**	**std err**	**t**	**p**	**CI (95%)**
Coefficient β_0 (intercept)	34.9467	2.498	13.992	0.000*	[29.187, 40.706]
Coefficient β_1 (indicator)	6.5244	1.771	3.683	0.006*	[2.439, 10.609]
Coefficient β_2 (AwardRatio)	1.9100	5.631	0.339	0.743	[-11.075, 14.895]
Coefficient β_3 (Authors)	-9.2715	5.969	-1.553	0.159	[-23.035, 4.492]
Coefficient β_4 (Arxiv)	-3.6121	1.651	-2.188	0.060	[-7.419, 0.195]
Coefficient β_5 (Repos)	-2.7982	29.674	-0.094	0.927	[-71.227, 65.631]
Coefficient β_6 (Reviews)	12.8483	48.702	0.264	0.799	[-99.459, 125.156]
Coefficient β_7 (Predators)	38.7569	10.727	3.613	0.007*	[14.021, 63.493]
Coefficient β_8 (Interaction)	3.3381	0.999	3.341	0.010*	[1.034, 5.642]
Coefficient β_t (Time)	1.8930	0.535	3.537	0.008*	[0.659, 3.127]

event study found that literature reviews do not have a statistically significant influence on the mean number of references. This can be explained by literature reviews, while growing in popularity, still being relatively rare in the CHI Proceedings.

(5) *An increase in the number of citations to arXiv pre-print articles* (Section 4.2.4):
Entire fields of science rely on arXiv as a means for article dissemination. The field of HCI is not immune to this trend, and it is becoming increasingly more common for CHI authors to cite unrefereed pre-prints. Before 2015, citations to arXiv were shunned upon by CHI authors. From 2016 onward, citations to arXiv pre-prints were no longer competing for space with citations to peer-reviewed articles. The decision to lift the page restrictions at CHI paved the way for this shift in citation behavior. Today, many CHI authors cite arXiv pre-print articles. This can also lead to adverse problems, such as pre-prints being cited instead of the published articles. While the trend of citing arXiv pre-prints was not statistically significant in the event study, the mean number of citations to pre-prints approaches significance ($p = 0.06$), hinting at a potential relationship that could develop significance in the future.

(6) *An increase in the number of citations to open data and code repositories* (Section 4.2.5):
Before 2015, there were virtually no formal citations to code or data repositories to be found in the reference sections of CHI articles. From our analysis, it can be concluded that lifting the page restrictions motivated authors to include more citations to such repositories in their CHI articles. From an economic lens, without a page limit, there no longer is a marginal cost to including formal citation to code and data repositories, and authors no longer have to trade-off between citations. The event study found this trend not to be statistically significant, which could be explained with the relative sparsity of citations to code and data repositories in the CHI Proceedings.

(7) *An increase in the number of citations to predatory journals and questionable publishers* (Section 4.2.6):
Without a trade-off cost between citations, authors are increasingly including citations to potential predatory journals and questionable publishers in their CHI articles. The long-tailed distribution of citations to publishers on Beall's List in Figure 14 highlights that the CHI community, overall, has not lost its sense for quality. The relative number of citations to smaller potential predatory journals and publishers is negligible compared to the overall number of citations in each proceedings year. However, the picture is different with larger questionable publishers (Frontiers Media, IOS Press, IGI Global, and IntechOpen) which are increasingly being cited by CHI authors. Perhaps this is yet another expression of the CHI community's obsession with quantity.

Together, these co-occurring trends shape the citation practices of the CHI community in the shifting landscape of HCI. The trends reflect an interesting gradual change of pace in academic working in the CHI community starting with CHI '16. The trends are also a reflection of broader changes that affect not only the field of HCI, but academia in general [10].

Like a frog slowly boiling in water, the CHI community is heating up under the relentless growth of references, risking a culture where quantity overshadows meaningful discourse and genuine innovation. A common proverb states: *"things need to get a lot worse before they get better"*. One has to wonder: At what point will the number of references per CHI article become unbearable for authors and peer reviewers – at 100 references, 130 references, 200 references? The pre-2016 limit on the number of pages in CHI articles constituted a "Schelling fence" [3] – a hard limit on the number of pages that could be included in CHI articles (including references). Authors could not cross this limit without risking a desk-reject of their paper. Thus, the space for references and content

had to be weighed against each other, and each additional reference reduced the space available for the article's content. Without these clear boundaries and trade-offs, the CHI community is on a slippery slope [3], a gradual erosion – or evolution, if one prefers to look at it this way – of norms and principles. The expansive path incentivizes authors to, year-by-year, value quantity over quality and include more references in their articles.

One could argue that the observed trend in Figure 1 is simply an effect of the overall growth of publications in the HCI field. However, while overall publication growth may certainly be a contributing factor, the growth of the CHI Proceedings cannot fully explain the observed growth in the mean number of references for two reasons. First, the growth of the CHI Conference Proceedings precedes the observed trend by 10 years, as discussed in Section 2.1. Second, the CHI Proceedings deviated from linear growth during the COVID-19 pandemic, while the mean number of references in CHI papers still increased steadily and fairly linearly. Therefore, the increase in the mean number of references per CHI article is likely not a mere side-effect of the proceedings growth at CHI.

What is striking about the observed trend in Figure 4 is that it follows a relatively stable linear growth pattern. This near-linear growth takes place in the presence of exponential growth in academia, such as the exponentially growing number of scholarly articles. Future work could investigate why the observed year-by-year growth in references is linear.

This near-linear growth could, perhaps, be explained with the simple psychological models of herd mentality and an availability heuristic. Each year, members of the CHI community observe — whether consciously or sub-consciously — an increase in the mean number of references per CHI article. In other fields of science, evidence has been found that longer articles are cited more [20]. Authors may, thus, decide to also include more references in their articles. Another explanation are the rising expectations of peer reviewers. The submission guide at CHI 2016 explicitly mentioned that *"lack of references to prior work is a frequent cause for complaint – and low rating – by reviewers"* [1]. It is not uncommon for peer reviewers to demand inclusion of a long list of references in their reviews. Authors often interpret these references not as suggestions but as mandatory for paper acceptance, and subsequently incorporate them into their paper. A simple strategy of "the more, the better" could be motivating authors in CHI to – preemptively – include more references in their articles to satisfy the growing expectations of peer reviewers.

Likely, though, it is a combination of several different co-occurring factors such as the ones presented in this article (i.e., the increase in systematic reviews in HCI, growing diversity of topics and co-authorship, growing expectations among peer reviewers, growing number of publications, and pressure to publish) that contributes to the increase in the mean number of references in CHI articles. We leave a detailed qualitative investigation of the motivation of authors to include more references in their papers to future work (for instance with interview studies, workshops, or a Delphi study). In the following section, we discuss why the growing number of references in articles is a critical concern that demands attention.

5.1 Implications for Authors, Reviewers, and the Broader Academic Community

References in scholarly articles serve an important function: they help novices in the field to learn and become familiar with the research area. However, we argue the observed trend (i.e., a culture of excessive citation, as evident in the rising number of references in CHI articles) is a negative development for authors, peer reviewers, the CHI community, and the entire field of HCI, for a number of different reasons.

First, we argue 130 references per article are too many references to thoroughly screen during the peer review process. A high number of references increases the cognitive load on peer reviewers, contributing to reviewer fatigue. This only adds to the already high amount of peer review fatigue, with conferences struggling to recruit high-quality reviewers. Peer review is the backbone of academic rigor, and if reviewers only give a cursory glance at the reference section, the integrity of the peer review process is compromised.

Second, the number of references included in CHI articles serves as a signal to the CHI community. Peer pressure and herd mentality suggest that if the average number of references is high, authors feel compelled to include a similar number of references in their articles. Deviations from this norm are likely to be penalized by peer reviewers, creating a spiraling pattern of year-by-year increasing academic pressure and competition. This spiraling pattern contributes to both author and peer reviewer dissatisfaction and fatigue.

Third, despite the shift away from awarding best paper awards and honorable mentions to high-reference articles, the overall rise in the mean number of references per article still sends the wrong message to junior members of the CHI community. The literature work required to publish an article at CHI poses a significant and undue burden on young scholars. Today, PhD students in HCI are faced with a seemingly insurmountable and overwhelming amount of literature, which distracts the students from developing their research skills and expertise.

Fourth, excessive citation contributes to inflation of the value of each citation. Some authors may use "citation stuffing" to prop up the perceived value of their articles. Due to this carelessness, a citation today does not carry the same weight it did in the past. This is similar to how early search engine optimization efforts devalued the significance of web links. The increasing number of citations reduces their impact and contributes to a hemorrhage of less meaningful references.

Last, the high number of references makes it easier for bad actors to commit misconduct and fraud. In an environment of hypercompetition and "perverse" incentives [10], academic fraud is surprisingly common [37]. With no clear trade-off cost between citations in an article, the reference section can be expanded arbitrarily, facilitating citation manipulation. This can include self-citation [6, 22], author collusion in citation cartels [16, 19], and even the purchasing of citations [11, 14]. The high number of references that, likely, undergo only cursory peer review makes it harder to detect such unethical and fraudulent research practices.

5.2 Reflective Considerations for Future Research Practices

Many of the developments in the CHI community are plan-driven. For instance, the number of articles accepted at the CHI Conference is deliberately expanded each year. One should not forget that such top-down decisions and policy changes lead to behavioral changes in members of the research community. However, while the CHI Steering Committee initially started the CHI community on its expansive path with its 2016 policy, it is the collective decision of authors at CHI to, year-by-year, cite more and more articles. In this section, we urge the community to pause and reflect on its citation practices.

This work is meant as a zealous vehicle for the members of the CHI research community to open their eyes to the community's growing problems. We presented trends in citation practices to foster dialogue — not to prescribe mandates, but to encourage diverse interpretations and applications of these findings. We invite readers to engage with the data, reflecting on its implications for the future of HCI research. Our work serves not as a directive, but as a mirror reflecting the current state of CHI's scholarly practices. We acknowledge that works like ours can be misinterpreted. We would like to explicitly stress that this work is not intended to be a "guide" or "recipe" for writing successful CHI papers. By offering a data-backed overview of growing pain points in the field of HCI, we instead aim to help scholars and practitioners navigate the complexities of the modern research landscape and drive positive change in the community.

However, a critical question arises. How much responsibility does each author in the CHI community bear for the observed changes in citation practices? While scientific knowledge advances rapidly, the academic system remains relatively conservative and resistant to change. This system, dominated by powerful publishers like ACM Press, often feels too vast for any one individual to influence significantly. Therefore, we also call for reflective consideration on an individual level. We urge CHI authors to be more deliberate with their citations, viewing them as quality signals and endorsements. Readers should consider how citation trends impact their research practices and the broader HCI community.

Through introspection, the community can shape future research in ways that are thoughtful and aligned with shared values. The introspective approach emphasizes the agency of the scholarly community to enact change, respecting the diversity of perspectives within our field and acknowledging each researcher's autonomy to interpret and integrate these insights according to their goals. We should acknowledge that, while fundamental goals (such as publishing articles in high-quality venues) are shared by all of us, the way to achieve these goals varies on an individual level, as demonstrated, for instance, by the 'slow science' movement [4]. This movement calls for a more deliberate and cautious approach to publishing. Moving forward, the CHI community should find ways to accommodate these diverse approaches in its citation practices.

5.3 Potential Solutions

In this section, we discuss potential solutions addressing the observed upward trend in the mean number of references in CHI articles. One potential response is to simply ignore the trend, though this approach risks future challenges. Proactively addressing the issue of excessive citations is a more constructive approach. This section discusses four potential solutions for addressing the growing number of references.

5.3.1 Solution 1. A first solution would be to revert to the 2015 policy (or an updated version thereof). This could involve setting a limit on the number of references, the number of reference pages, or the overall number of pages per article, as was the case in 2015. At that time, a citation was an endorsement, and any citation had to be weighed against the space it would take away from the article's content or other references. This trade-off has since been lost.

While modern research practice tends to allow an unrestricted number of pages and references, recent iterations of the CHI Conference have seen strong nudging toward lower page limits. For instance, recent CHI conferences clearly signaled that unrestricted growth in article length is not welcomed by the Conference Chairs and the Steering Committee. The suggested number of words for CHI articles in 2024 was 7000–8000 words – far lower than the average [29]. Yet, the strong recommendations are inconsistently enforced by the subcommittees. Limiting the number of references for most articles (with the exception of literature reviews) could be a viable solution to address the growing and excessive number of references included in CHI articles. For such an approach to be successful, hard limits would need to be set, without exceptions, and these limits would need to be consistently enforced by all subcommittees.

5.3.2 Solution 2. A second solution is to develop language models for supporting paper writing. In particular, specialized language models could be trained to automatically cite prior work, given the text of a manuscript. From the perspective of the author, this would alleviate much of the literature work, and it would help authors concentrate on ideation and conducting high-quality work. However, as the failed release of Meta's Galactica model demonstrated in 2022 [13], language models are still prone to hallucinate non-factual and biased information which makes these models, in their present form, not suitable for literature work. This solution would also not alleviate peer reviewer fatigue, which we discuss in the next section.

5.3.3 Solution 3. A third solution pertains to the nature of the peer review process in academia. Peer review is the cornerstone of scholarly rigor, serving as a crucial filter that ensures the quality, relevance, and integrity of academic publications. From the perspective of creativity theory [33] and Csikszentmihalyi's systems model of creativity [9], the peer review process is a gatekeeping process that validates two critical components of an article: novelty and importance (also referred to as usefulness or appropriateness) [33].

As gatekeeper, it is the peer reviewer's task to assess these two components and, hence, the creativity of a submitted article.

In some cases, an extensive related work section may support the evaluation of an article's novelty, especially in cases where the peer reviewer may not have sufficient knowledge for conducting the review. But in an ideal world, a peer reviewer lacking sufficient knowledge of the field should not review an article in the first place. In practice, however, a mismatch between reviewer expertise

and a paper is, unfortunately, rather common [34], especially in large and highly diverse conferences, such as CHI. In the role of a gatekeeper, it is the peer reviewer's duty to be knowledgeable about works in the field. In a field as vast and diverse as HCI, this can be challenging.

A potential solution is to split peer review into two parts, one conducted by the human peer reviewer, the other one performed by a machine. Language models are knowledge compressors and can assess whether a contribution is original and whether the references are comprehensive more effectively than human reviewers. This would allow human reviewers to focus on assessing the second component of creativity: the importance of the methodologies, findings, and contributions for the field.

Evidence suggests we are already moving toward this solution. In the United States, automated essay scoring systems are already grading student essays [38] and a growing number of peer reviewers use language models to write their reviews. A study by Liang et al. found that up to 17% of peer reviews at top machine learning conferences are substantially authored by language models [24]. Liang et al.'s study found an overlap of between 30.9% and 39.2% between the points raised by GPT-4 and human reviewers, and their user study with 308 researchers from 110 institutions in the US highlighted that researchers find AI-generated feedback useful. Similarly, Sun et al. found that automated peer review significantly expedited the review process and improved review coverage, although raising concerns around trust, bias, and agency [36].

Automating aspects of peer review could offer significant benefits [21]. The cost of peer review is estimated to be billions of dollars annually [2]. In a system strained by an overwhelming volume of literature [12], partial automation can alleviate the burden on reviewers. Further, automating peer review could increase trust in the scientific process by reducing biases, such as the Matthew effect, which has been a concern in academia since at least 1968 [27].

In the future, language models could assist in evaluating novelty, leaving other aspects to human reviewers, thereby enhancing the efficiency and reliability of the peer review process.

5.3.4 Solution 4. The potential solutions discussed so far have centered around policies and peer review. Another idea focuses on driving change from the author side. In some disciplines, journals require authors to pre-register their studies before conducting them. What if CHI authors pledged to submit their articles with a predetermined number of references? Authors could set this limit for themselves and they would pre-register this limit publicly before writing and submitting their paper. The pre-registered limit would work as a self-binding contract, forcing authors to keep within their pledged number of references. This – like in the years before the page limit was lifted – would force authors to make a trade-off between references and to think more carefully how many and which references to include in their articles. To incentivize participation in this pre-registration scheme, awards could be given to authors who conform to their pre-registered number of references. The public dataset of pre-registrations could contribute to establishing new community norms and standards for the number of references in CHI articles. However, these standards would not need to be the same for every author. For instance, students could be allowed to include fewer references, acknowledging their role as junior members in the research community. This would alleviate the undue burden of literature work on young scholars.

6 Conclusion

This article presented meta-research on citation practices at the ACM CHI Conference, the top conference in the field of Human-Computer Interaction. Our exploratory analysis provided clear evidence of a change of pace in academic working in the CHI community, indicative of broader trends in HCI. We provided empirical evidence of a year-by-year increase in the number of references included in CHI articles. The excessive citation practices were enabled by a policy decision that lifted page restrictions on the reference section of CHI articles. This policy decision constitutes a non-linearity in the CHI community's trajectory and had a destabilizing effect on the community. The implications of this meso-level policy decision are profound, underlining the value of meta-research for the CHI community and HCI. More meta-research is needed to carefully consider the broader impact of policy decisions and to provide clear recommendations for stakeholders. It is time for the HCI community to pause and reflect on its citation practices and constructively discuss solutions. A different pathway forward is urgently needed.

Acknowledgments

The author aimed to limit this paper to 35 references, but unfortunately, this target was exceeded. I pledge to do better next time.

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